Ray Pomponio

Pittsburgh, PA • github.com/rpomponio • ncbi.nlm.nih.gov/myncbi/raymond.pomponio.1

Education

MS, Biostatistics May 2023
University of Colorado–Denver
Thesis: *Mistaken Identities Lead to Missed Opportunities: Testing for mean difference in partially matched data*
Marvin Porter Award for Outstanding Academic Achievement

Graduate Coursework, University of Pittsburgh December 2025
Introduction to Causal Inference (graduate-level biostatistics)

BS, Economics May 2018
University of Pennsylvania, Wharton School
Dean's List

Technical Skills

Programming: Python (Matplotlib, NumPy, Scikit-learn, SciPy, PyTorch), R (tidyverse, mice, data.table), SAS, Unix shell scripting

Machine Learning & AI: Deep learning (CNNs, brain age modeling), supervised/unsupervised ML, Monte Carlo simulation, cross-validation, model selection, image processing, computer vision

Statistics: Multivariate regression, Bayesian data analysis, causal inference, propensity scores, survival analysis, longitudinal data, missing data/multiple imputation, sample size/power calculation

Data Infrastructure: SQL, DBT, Git-based workflows, REDCap (including API automation), Microsoft Power BI, Power Automate, end-to-end clinical data pipeline design

AI-Assisted Workflows: Claude (Anthropic), GitHub Copilot; applied to scientific report drafting, slide narrative development, code generation and debugging, literature synthesis, and translating analytical outputs into non-technical communications

Operating Systems: Linux, macOS, high-performance computing environments (HPCs)

Research & Professional Experience

Data Scientist 2024 – Present
University of Pittsburgh / Children's Hospital of Pittsburgh

- Lead biostatistician on a **CDC-sponsored New Vaccine Surveillance Network** study tracking respiratory viruses across 8,000+ pediatric encounters; designed and executed the full clinical analytics pipeline from study design through publication in *mBio* (2026).
- Collaborated directly with **CDC program scientists** on study design, data governance, and public health interpretation, translating policy-relevant questions into epidemiological data collection instruments.
- Designed and managed end-to-end data infrastructure spanning multiple clinical systems (REDCap, EHR extracts, laboratory databases), enabling greater security and faster time-to-analysis in routine reporting.
- Collaborated with clinical investigators to define research questions, then developed and vali-

dated **multivariate machine learning risk-factor models** for predicting severe respiratory illness in pediatric populations.

- Mentored junior faculty, fellows, and MPH student on statistical methods for infectious disease research, leading to 3 successful proposals, one thesis defense, and one K-award submission.
- Integrated LLM-assisted workflows (Claude, GitHub Copilot) into routine deliverables, including automated report drafting, slide deck narratives for CDC stakeholders, and code generation, accelerating turnaround on recurring analytic tasks.

Independent Statistical Consultant 2023 – 2024
Upwork.com

- Advised medical device clients on **FDA submission study design**; performed sample size and power calculations to justify endpoints and regulatory proposals.
- Conducted **customer segmentation analysis** for an e-commerce client using clustering methods, identifying high-value repeat-purchasing behaviors to inform retention strategy.
- Used AI-assisted tools (Claude, GitHub Copilot) to streamline client deliverables, including synthesizing regulatory literature and generating clear written summaries of technical findings for non-specialist audiences.

Graduate Research Assistant 2021 – 2023
University of Colorado–Denver, Department of Biostatistics & Informatics

- Co-authored **10+ peer-reviewed manuscripts** and conference abstracts across pulmonary medicine, critical care, and clinical epidemiology, spanning seven journals.
- Collaborated with multidisciplinary teams of clinicians, epidemiologists, and pulmonary/critical care specialists to translate clinical questions into testable statistical hypotheses and funding applications.
- Completed doctoral-level coursework in **machine learning and model selection**.

Data Analyst 2018 – 2020
University of Pennsylvania, School of Medicine
Davatzikos Lab (Center for Biomedical Image Computing and Analytics)

- Developed **neuroHarmonize**, an open-source Python package for harmonizing neuroimaging datasets across 20+ medical centers using ComBat-based statistical methods; now with 100+ GitHub stars and **500+ citations**.
- Built automated **machine learning pipelines for image processing and feature extraction** that reduced MRI preprocessing timelines from months to days for a multi-center consortium studying biomarkers of neurodegeneration.
- Applied **deep learning and computer vision methods** to large-scale MRI datasets (14,000+ individuals) to model brain age trajectories and disentangle neurodegeneration from typical aging patterns.
- Presented novel methodologies and technical results to interdisciplinary audiences including neurologists, radiologists, and biostatisticians at Penn, UCSF, and the iSTAGING consortium.

Grants & Funding

- New Vaccine Surveillance Network, Centers for Disease Control and Prevention. Role: Data Scientist/Statistician. PI: Dr. Marian Michaels. University of Pittsburgh, 2024–present.

Awards & Honors

- Certificate of Appreciation. Centers for Disease Control and Prevention, 2024.
- Marvin Porter Award for Outstanding Academic Achievement. University of Colorado–Denver, Department of Biostatistics & Informatics, 2023.
- Dean's List. University of Pennsylvania, Wharton School, 2018.
- Hackathon Winner. Ryder Transportation customer analytics, Wharton Customer Analytics Initiative, 2018.

Selected Publications

Full list: ncbi.nlm.nih.gov/myncbi/raymond.pomponio.1/bibliography/public/

Neuroimaging & Harmonization

- **Pomponio R**, Erus G, Habes M, Doshi J, Srinivasan D, Mamourian E, et al. Harmonization of large MRI datasets for the analysis of brain imaging patterns throughout the lifespan. *NeuroImage.* 2020;208:116450.
- Bashyam VM, Erus G, Doshi J, . . . , **Pomponio R**, et al. MRI signatures of brain age and disease over the lifespan based on a deep brain network and 14,468 individuals worldwide. *Brain.* 2020;143(7):2312–2324.
- Habes M, **Pomponio R**, Shou H, et al. The Brain Chart of Aging: machine-learning analytics reveals links between brain aging, white matter disease, amyloid burden, and cognition in the iSTAGING consortium of 10,216 harmonized MR scans. *Alzheimer's & Dementia.* 2021;17(1):89–102.
- Nasrallah IM, Gaussoin SA, **Pomponio R**, et al. Association of intensive vs standard blood pressure control with MRI biomarkers of Alzheimer disease: secondary analysis of the SPRINT MIND randomized trial. *JAMA Neurology.* 2021;78(5):568–577.
- Chen AA, Srinivasan D, **Pomponio R**, et al. Harmonizing functional connectivity reduces scanner effects in community detection. *NeuroImage.* 2022;256:119198.

Clinical & Epidemiological Research

- **Pomponio R**, Peterson RA, Owusu M, et al. Phosphatidylethanol measures in patients with severe COVID-19-associated respiratory failure identify a subset with alcohol misuse. *Alcoholism: Clinical and Experimental Research.* 2025;49(1):165–174.
- Pless LL, Damodaran L, **Pomponio R**, et al. Emergence and epidemiology of dominant variants of human metapneumovirus in the United States between 2016 and 2021. *mBio.* 2026;17(2):e0261925.
- Dunbar PJ, Peterson R, McGrath M, **Pomponio R**, et al. Analgesia and sedation use during noninvasive ventilation for acute respiratory failure. *Critical Care Medicine.* 2024;52(7):1043–1053.
- Smith JB, Peterson RA, **Pomponio R**, et al. Donor derived cell free DNA in lung transplant recipients rises in setting of allograft instability. *Frontiers in Transplantation.* 2024;3:1497374.

- Mayer M, Badesch DB, . . . , **Pomponio R**, et al. Impact of the COVID-19 pandemic on chronic disease management and patient reported outcomes in patients with pulmonary hypertension. *Pulmonary Circulation.* 2023;13(2):e12233.

Presentations & Teaching

- Presented thesis research to the Colorado School of Public Health, 2023.
- Lightning talk and poster presentation, Symposium on Data Science and Statistics, 2022.
- Seminar presentations on methods in biostatistics and data science, 3+ venues.
- Teaching Assistant, Longitudinal and Survival Analysis, University of Colorado–Denver, 2023. Led 10+ interactive office hours.
- Referee, *Imaging Neuroscience* and *NeuroImage* (5+ manuscripts reviewed).